

Software Modeling and UML

Experiment 3

Class Diagram Modeling

Objectives

- Based on the use case models constructed in the previous experiments, conduct the class diagram modeling and provide the basis for further design.
- Become familiar with the general process of modeling class diagrams.
- Gain more familiarity with the use of UML tools and master the representation of common modeling elements in class diagrams.

Case one: Library management system (Personal work)

- Become familiar with the representation and usage of modeling elements and their relationships in class diagrams.
- Based on the use case model of **Library management system** constructed in the previous experiment, draw class diagrams for the **Library management system**.

Case two: Human resource management system (Personal work)

- Become familiar with the representation and usage of modeling elements and their relationships in class diagrams.
- Based on the use case model of the Human resource management system constructed in the previous experiment, draw class diagrams for the Human resource management system.

Case three: Research project management system (Personal work)

- Be familiar with the representation and use of modeling elements and their relationships in class diagrams.
- Based on the use case model of **Research project management system** constructed in the previous experiment, draw class diagrams for the **Research project management system**.

Experiment requirements

- Please send your work through the link provided by the teaching assistant's cloud storage.
- Requirements for the report
 - Report name: Student ID Name UML EX3 Class diagram + case name
 - Please do not compress any document (不要压缩)
 - The deadline for task submission is 6 PM this Saturday evening
- Make sure to send the report following the specified guidelines; otherwise, we will not be able to receive your submission.

Notes

- Understand the relationships between classes during the class diagram modeling process.
- Experience the distinctions between the three perspectives of class diagrams during the modeling process.
- Determining classes is a critical and challenging task that requires careful consideration throughout the modeling process.
- Completing the class diagram modeling is just one outcome. It's crucial to grasp the intricacies of the process; the journey is more important than the end result.

Please consider the following questions

- In what views can class diagrams be created?
- What stereotypes for classes are supported in the UML tools?
- What types of relationships exist between classes? Can these relationships be created using the UML tools?
- Is an actor (in use case modeling) a type of class?

How To Find Classes

- Identify candidate classes from the nouns in the use case description.
- Utilize the CRC (Class Responsibility Collaborator) method to identify classes.
- Apply the division of Boundary, Control, and Entity Classes.
- Determine classes based on design patterns.
- Discover classes following the software development process guidelines.

Steps Of Class Diagram Modeling

- Conduct domain analysis to establish system requirements.
- Identify classes and their responsibilities, determine attributes and operations.
- Preliminarily establish relationships between classes.
- Refine the relationships between classes.
- Create the class diagram and provide the necessary textual description.